

Systems Genetics

Spring 2026

Introduction to systems genetics

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Overview of course

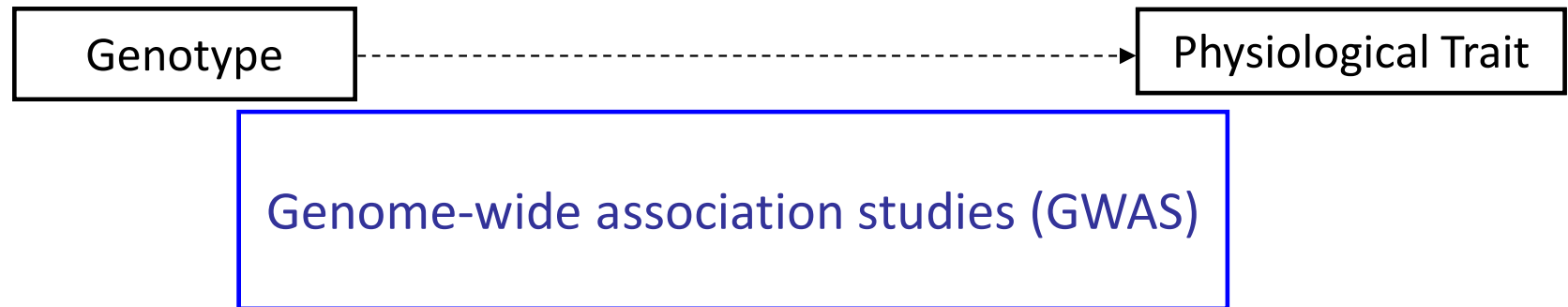
- Continuous and discrete traits, genetic basis
- The genomic era, GWAS
- Polygenic risk scores
- Systems genetics (this lecture)
- Next topics:
 - Causal inference ← BN material Project
 - Controlling for covariates

Main issues

- The rationale of systems genetics
- eQTL analysis - definition and rationale
- Common features of eQTL studies
- eQTL applications

The rationale of systems genetics

Poor results (e.g., missing heritability) for a long time



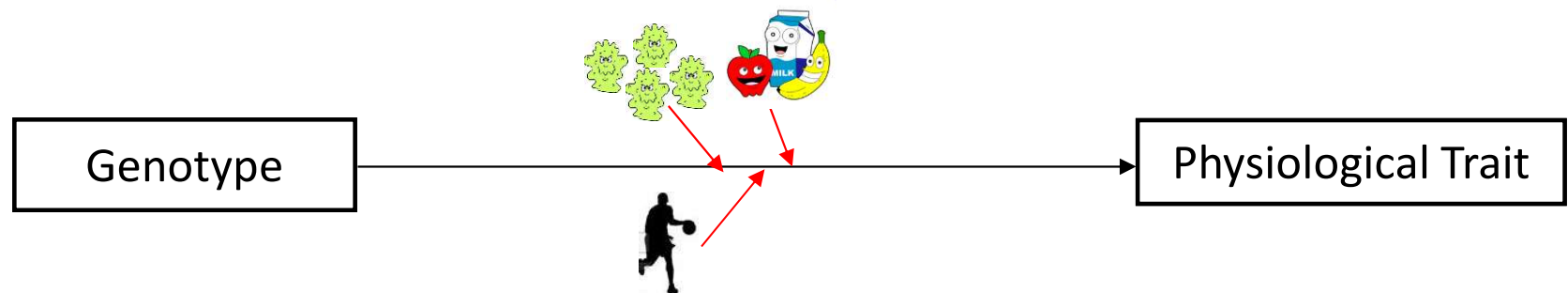
Poor results (e.g., missing heritability) for a long time why?



1. Indirect effect of genetics on trait

Poor results for a long time

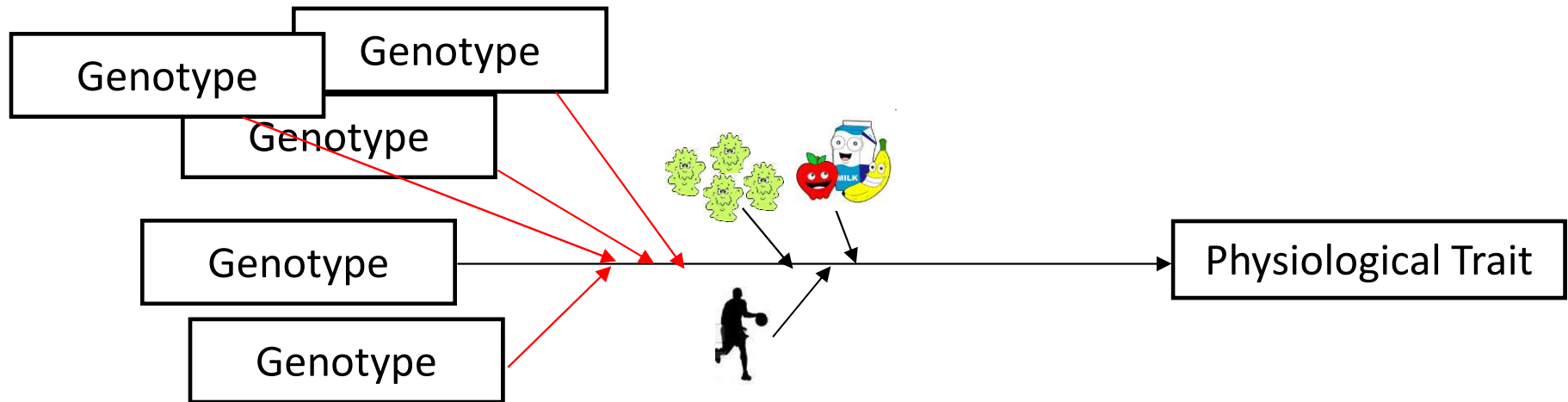
why?



1. Indirect effect of genetics on trait
2. Environmental variation

Poor results for a long time

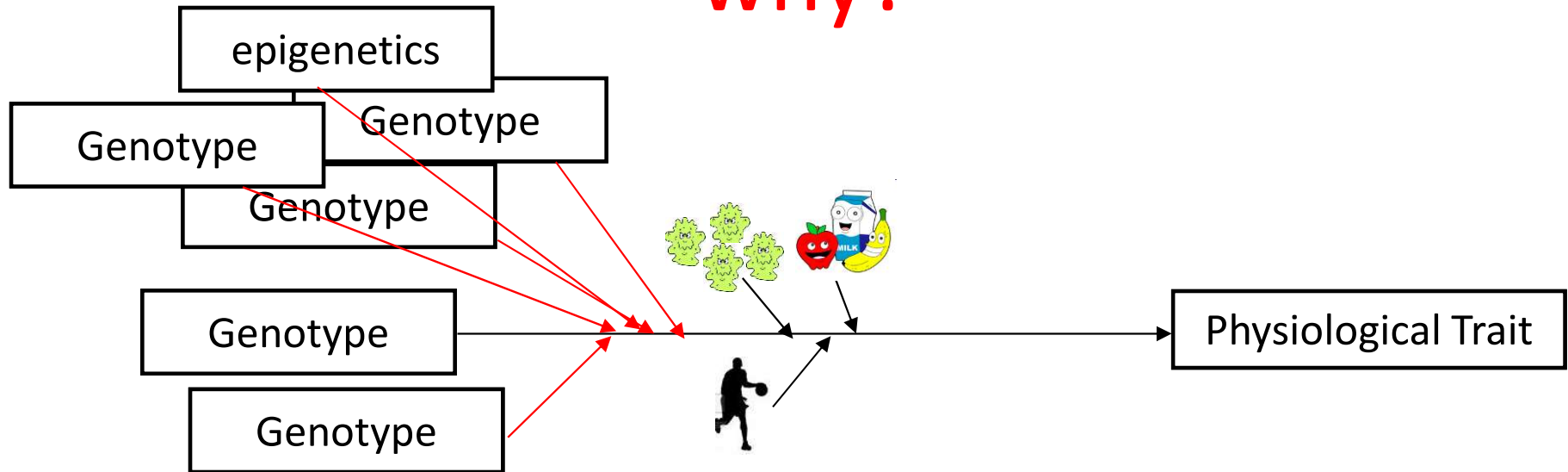
why?



1. Indirect effect of genetics on trait
2. Environmental variation
3. Multiple genetic variants underlying trait

Poor results for a long time

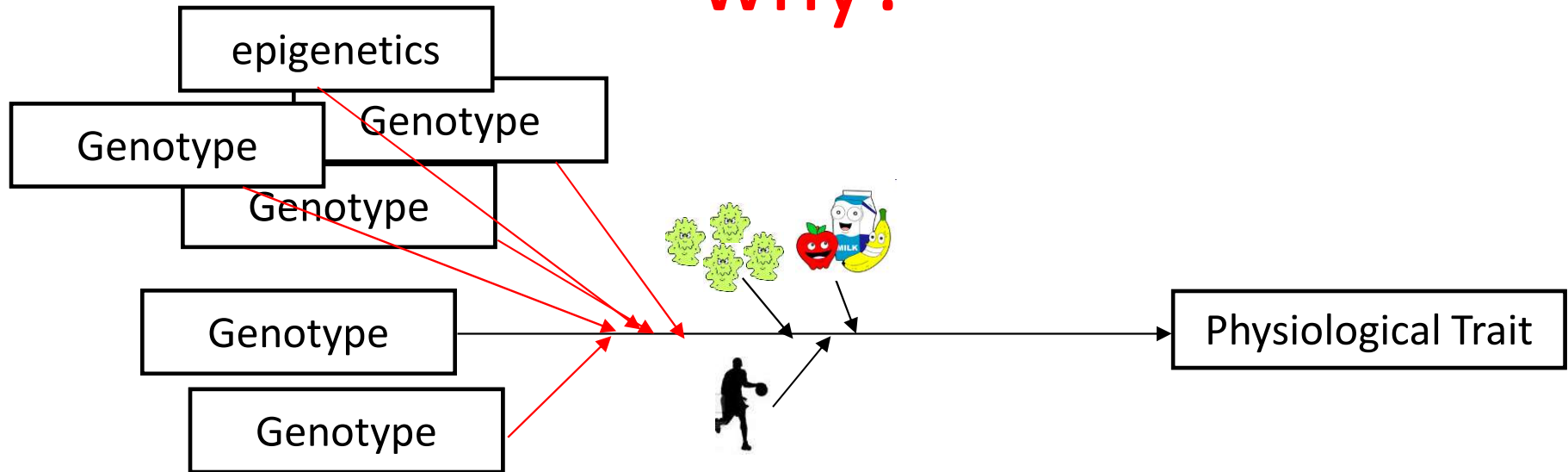
why?



1. Indirect effect of genetics on trait
2. Environmental variation
3. Multiple genetic variants underlying trait
4. Epigenetic variation

Poor results for a long time

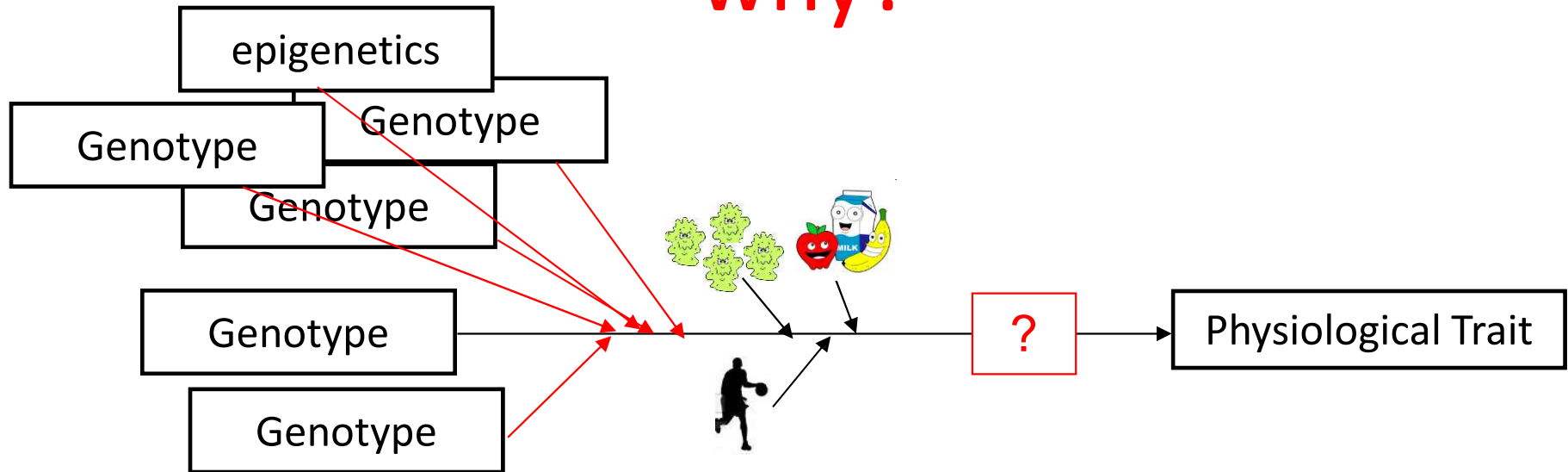
why?



1. Indirect effect of genetics on trait
2. Environmental variation
3. Multiple genetic variants underlying trait
4. Epigenetic variation
5. Lack of statistical power due to multiple testing

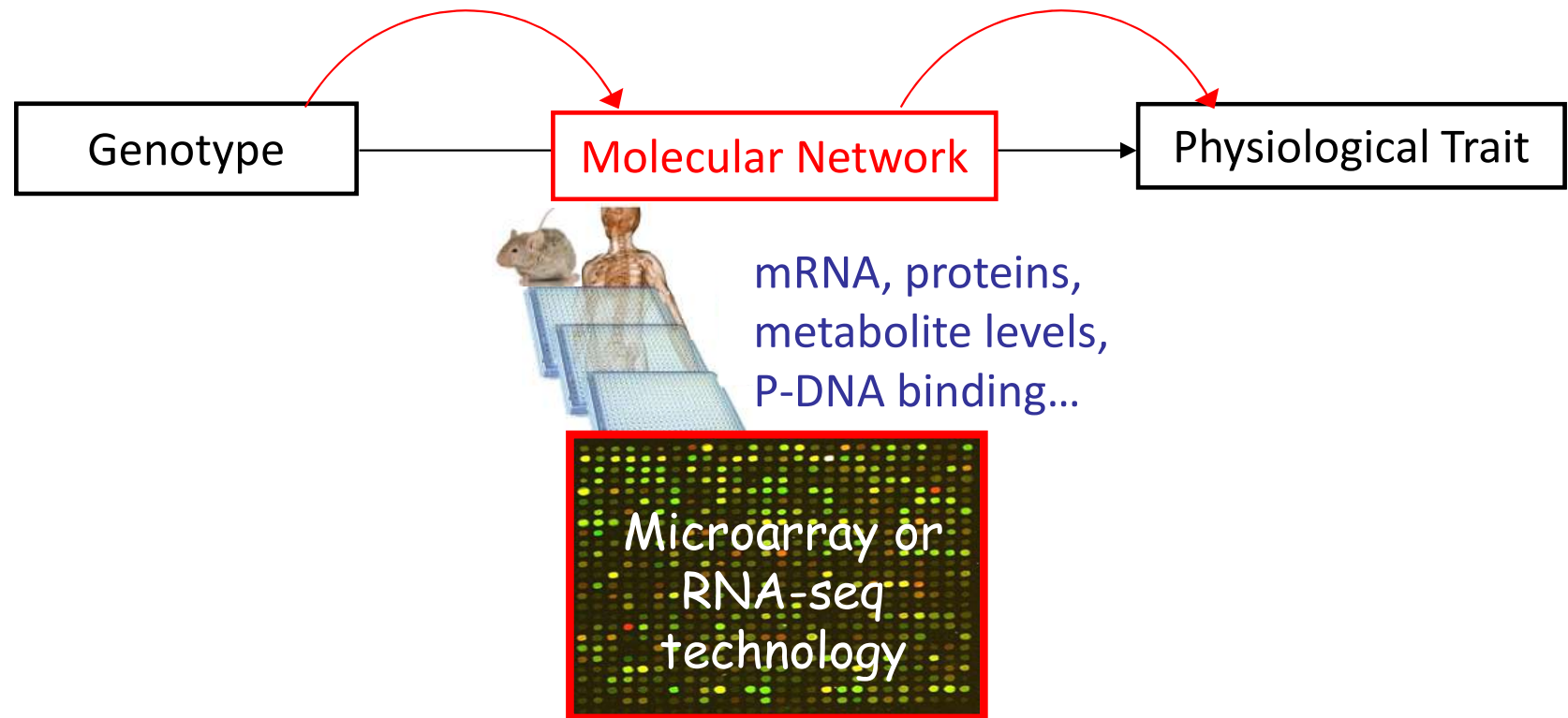
Poor results for a long time

why?

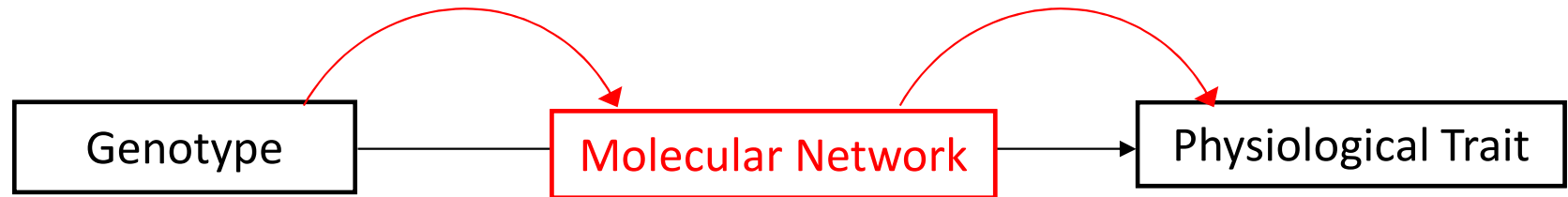


1. Indirect effect of genetics on trait
2. Environmental variation
3. Multiple genetic variants underlying trait
4. Epigenetic variation
5. Lack of statistical power due to multiple testing
6. Disease mechanisms remain unidentified

A strategy to predict a susceptibility trait

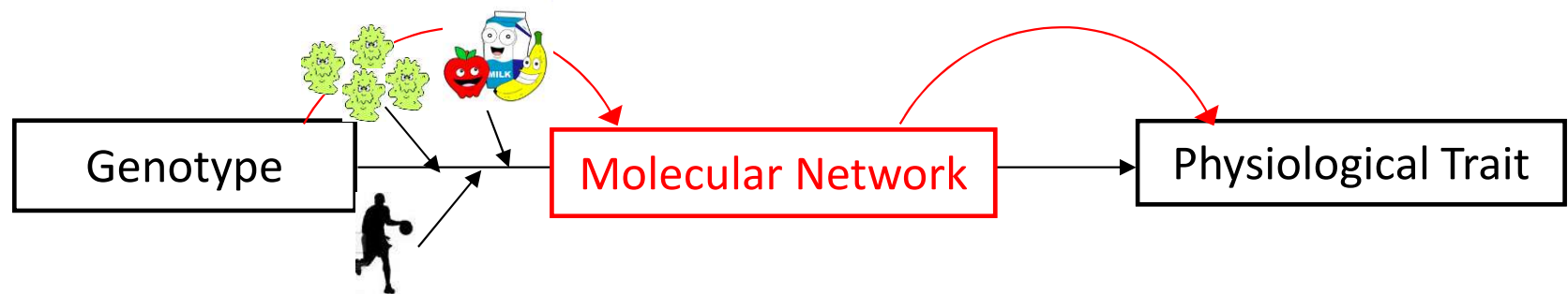


A strategy to predict a susceptibility trait



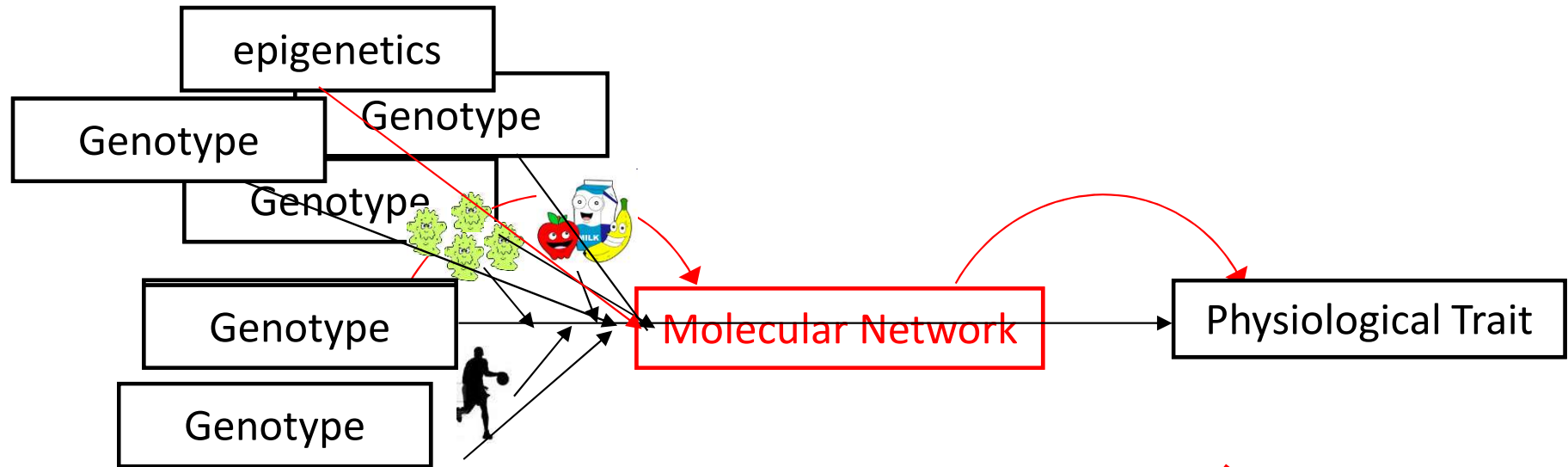
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A strategy to predict a susceptibility trait



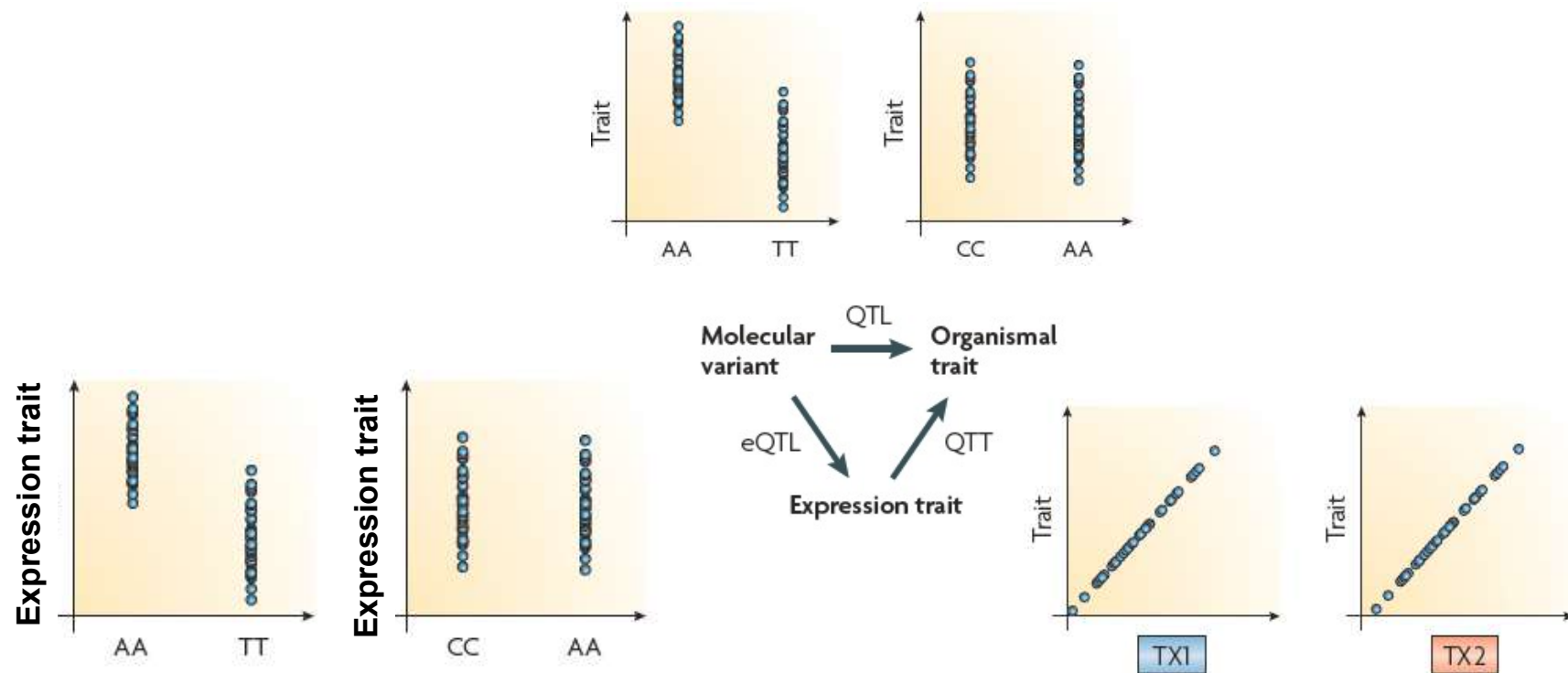
1. Indirect effect of genetics on trait
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A strategy to predict a susceptibility trait



1. Indirect effect of genetics on trait ✓
2. Environmental variation ✓
3. Multiple genetic variants underlying trait ✓
4. Epigenetic variation ✓
5. Lack of statistical power due to multiple testing ✓
6. Disease mechanisms remain unidentified ✓

Combining **gene expression** in a systems genetics framework



eQTL - definition and rationale

What is eQTL mapping?

- ▶ **Quantitative traits:** as opposed to dichotomous traits, have continuous trait values (eg, plasma cholesterol).
- ▶ **Quantitative trait locus:** a chromosomal region that affects the levels of a heritable quantitative trait.
- ▶ QTL mapping methods have been developed to identify potential QTL for a specific phenotype trait via either linkage information or association information, or both.
- ▶ Gene expression—mRNA abundance of a given gene—is also heritable.
- ▶ eQTL (**e**xpression **Q**uantitative **T**rait **L**ocus) is then the regulatory region for an expression trait.
- ▶ eQTL mapping is then QTL mapping with an expression trait used as the phenotype.

Focusing on eQTLs – why?

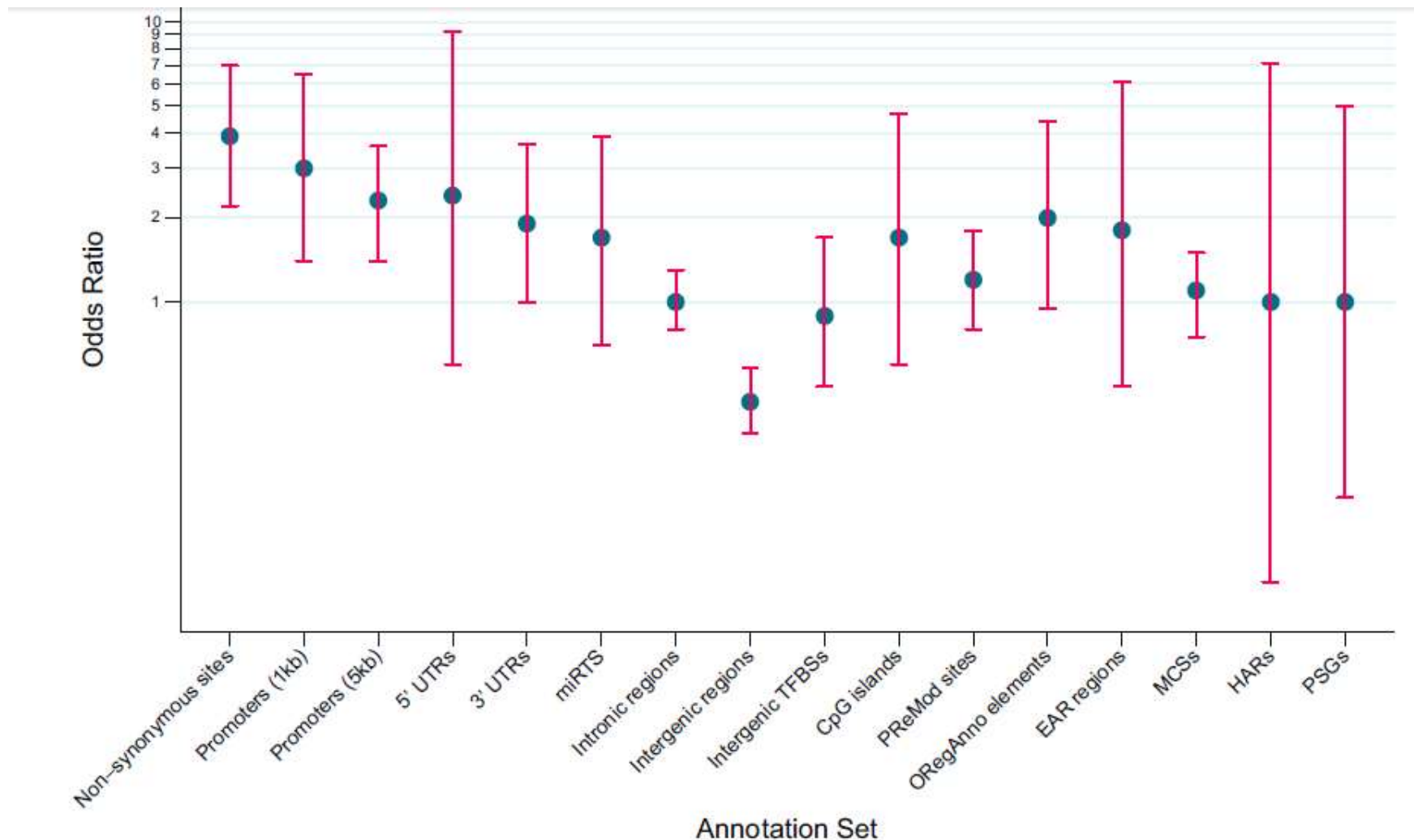
1. Over-representation of associated variants in promoters

Annotation set	Description	Source	Frequency (n, %) of TAS blocks*	Frequency of Random LD blocks*
Non-synonymous	Genomic positions wherein a nucleotide substitution would cause an amino acid replacement	dbSNP version 129	57, 12.2%	15.8, 3.5%
Promoters (1kb)	1kb regions upstream of annotated transcription start sites	Ensembl v49 release at www.ensembl.org	26, 5.5%	8.8, 1.9%
Promoters (5kb)	5kb regions upstream of annotated transcription start sites	Ensembl v49 release at www.ensembl.org	65, 13.9%	30.3, 6.7%
5' UTR	5' untranslated regions	dbSNP version 129	7, 1.5%	2.8, 0.6%
3' UTR	3' untranslated regions	dbSNP version 129	28, 6.0%	14.6, 3.2%
Intronic	Non-coding regions within a gene	dbSNP version 129	189, 40.3%	183.2, 40.3%
Splice sites	Intronic regions that allow for splicing (removing introns and joining exons)	dbSNP version 129	0, 0%	0.1, 0.02%
Intergenic	Non-coding regions outside of genes	dbSNP version 129	186, 40.0%	276.0, 60.7%

Based on a complete catalogue of SNP-trait associations in human

Focusing on eQTLs – why?

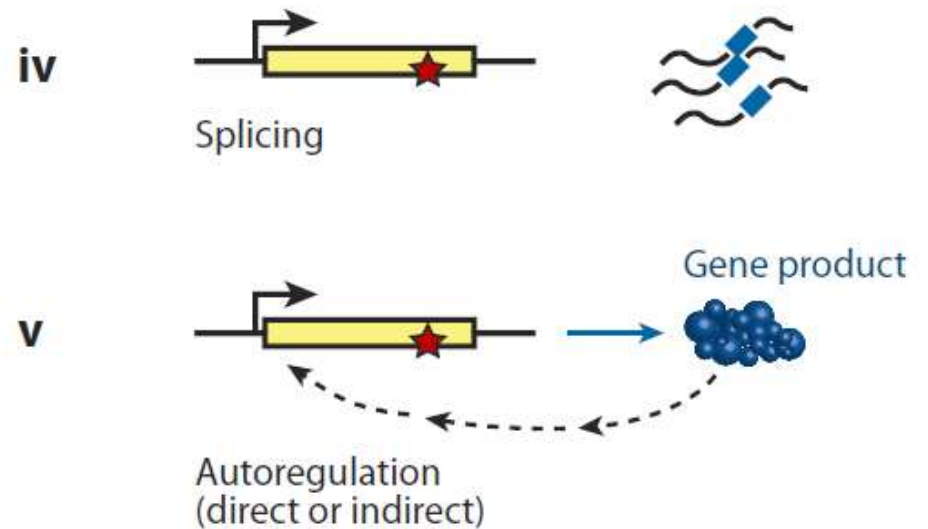
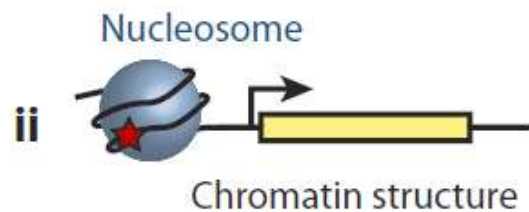
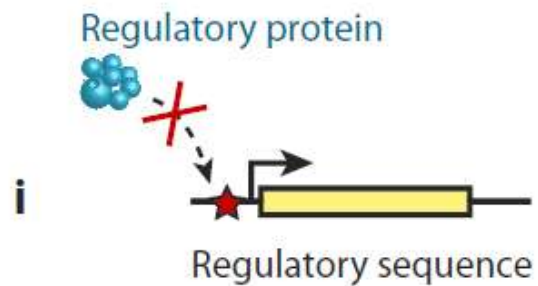
2. Associated variants in promoters tend to high effect size



Based on a complete catalogue of SNP-trait associations in human

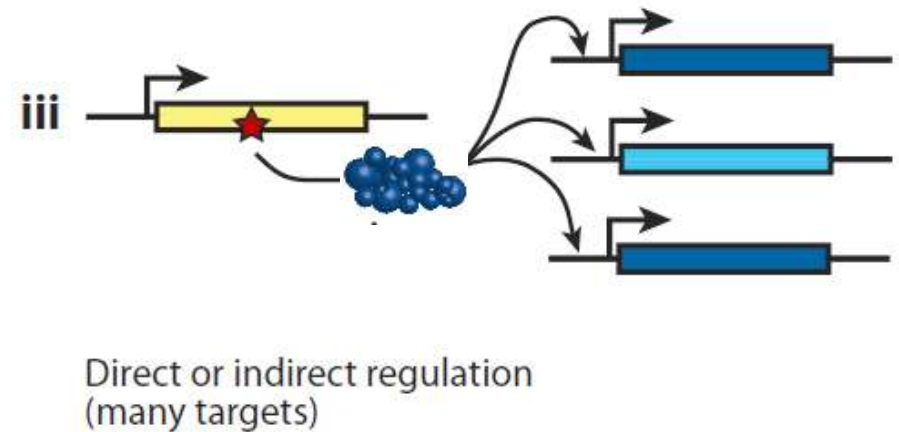
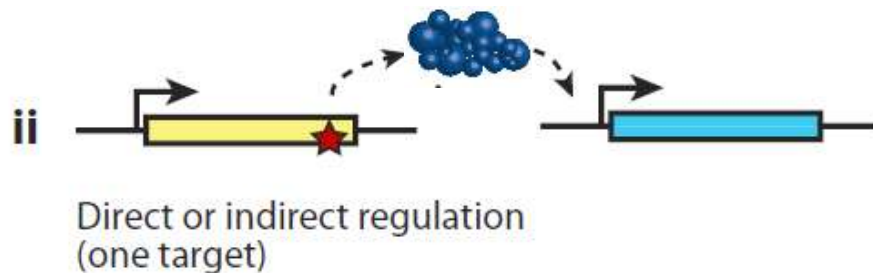
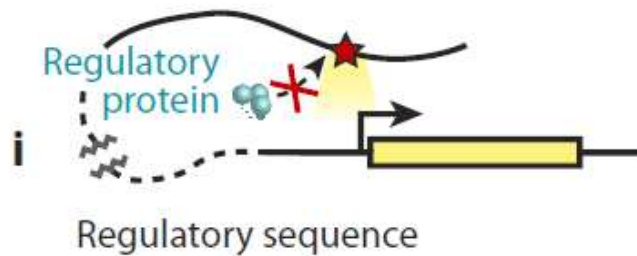
A MULTITUDE OF MECHANISMS CONTRIBUTE TO EXPRESSION VARIATION

a Local



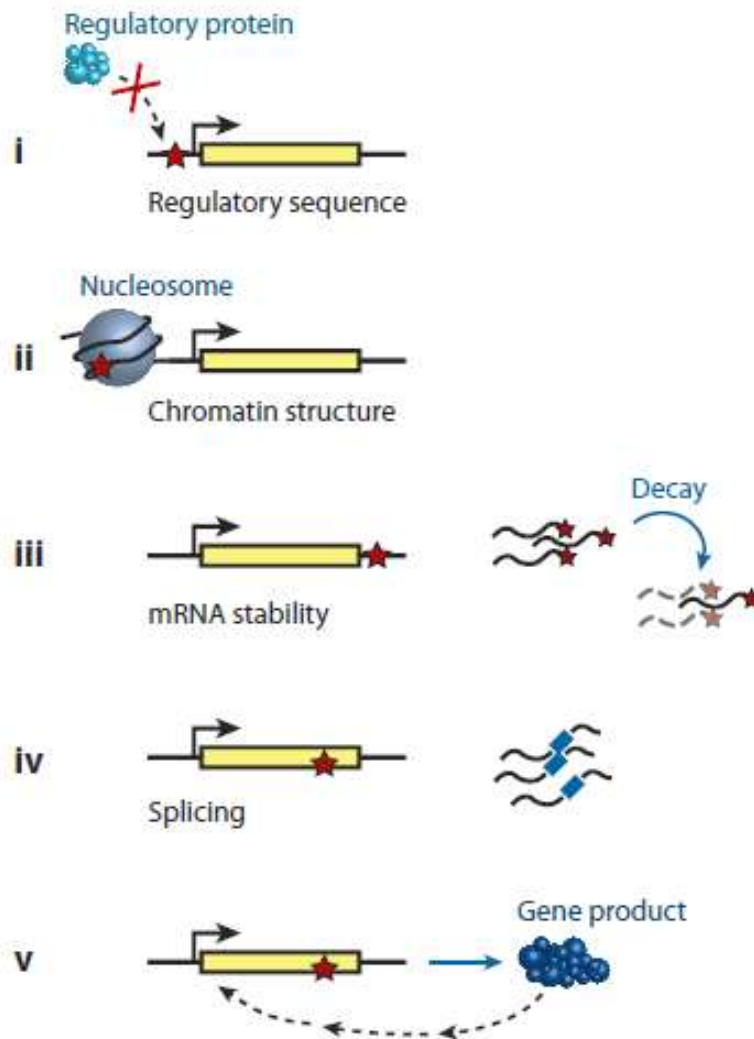
A MULTITUDE OF MECHANISMS CONTRIBUTE TO EXPRESSION VARIATION

b Distant



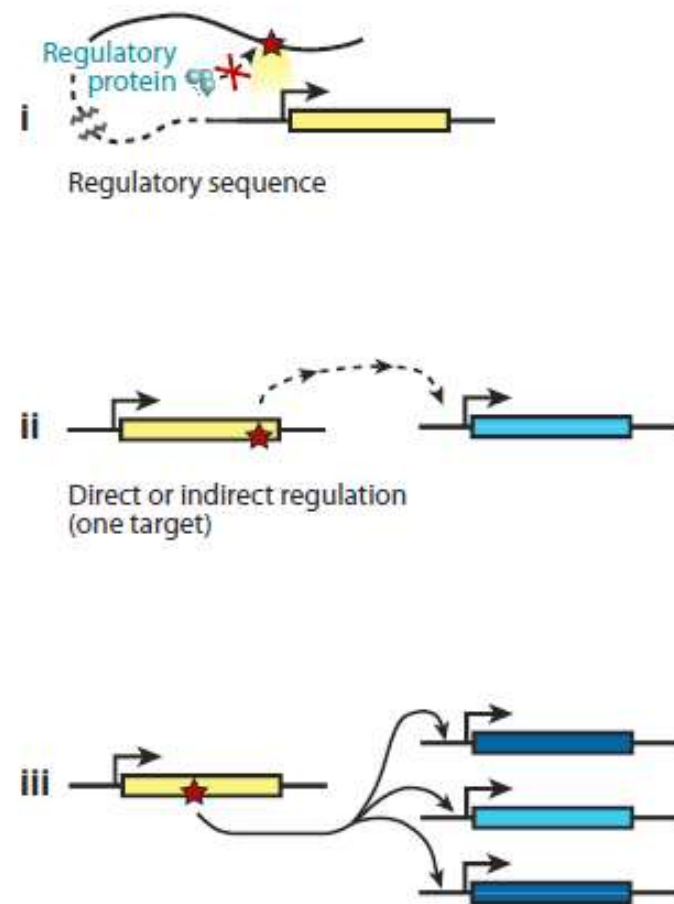
A MULTITUDE OF MECHANISMS CONTRIBUTE TO EXPRESSION VARIATION

a Local



Cis-acting eQTL

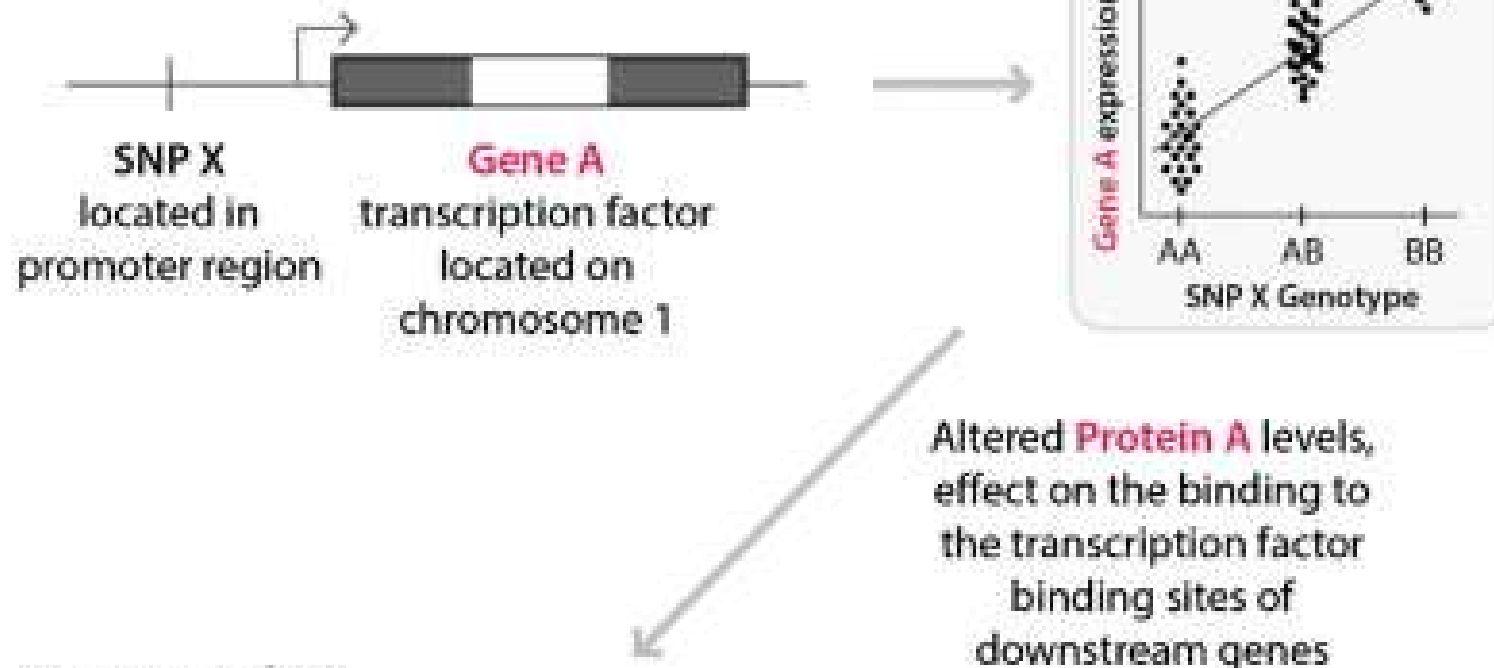
b Distant



Trans-acting eQTL

Cis-eQTL

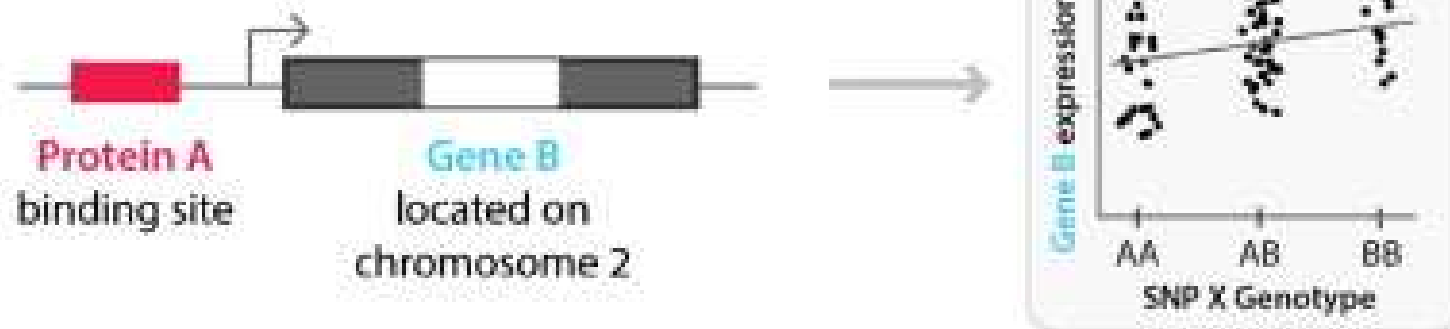
SNP X has an effect on local Gene A



Altered **Protein A** levels, effect on the binding to the transcription factor binding sites of downstream genes

Trans-eQTL

SNP X has an effect on distant Gene B through an intermediary factor (such as a transcription factor)



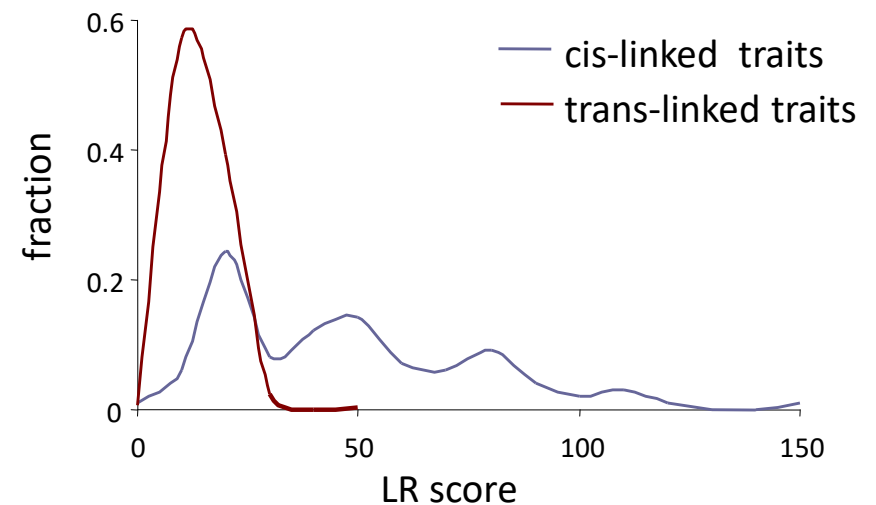
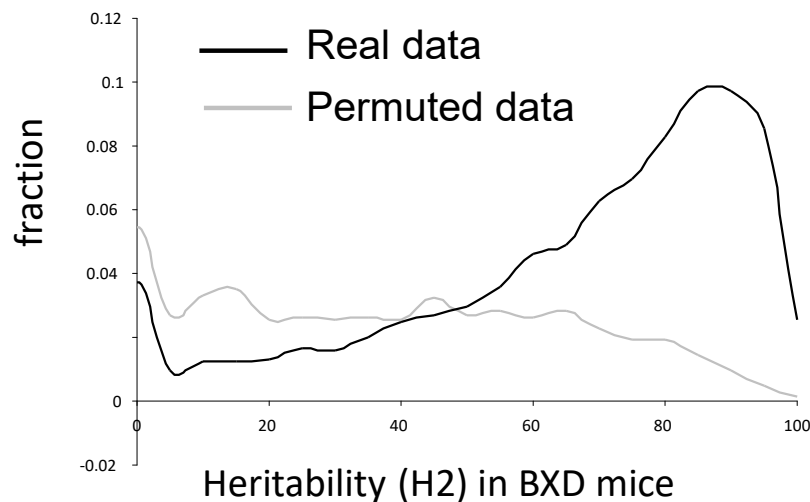
eQTL vs. QTL studies

1. The number of traits, that is, transcript levels, tends to be much larger than the number of individuals in the study.
2. transcripts have a local genomic context.
 - If the molecular variant is located within the gene region of the transcript under investigation, the regulation is called a *cis*, proximal or local eQTL
 - if the polymorphism associated with variation in the transcript is in another gene, it is called a *trans* or distal eQTL

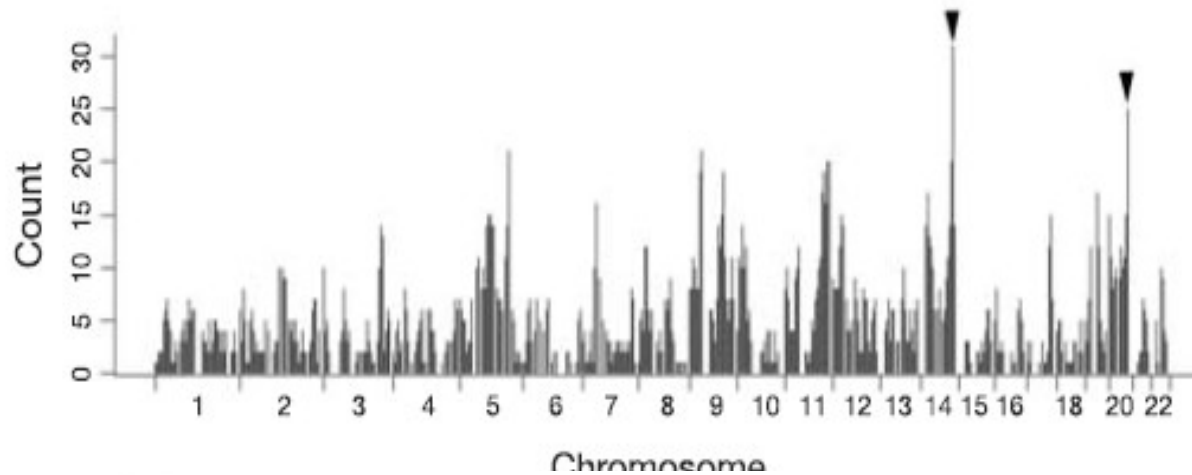
Common features of eQTL studies

Common features of eQTL studies

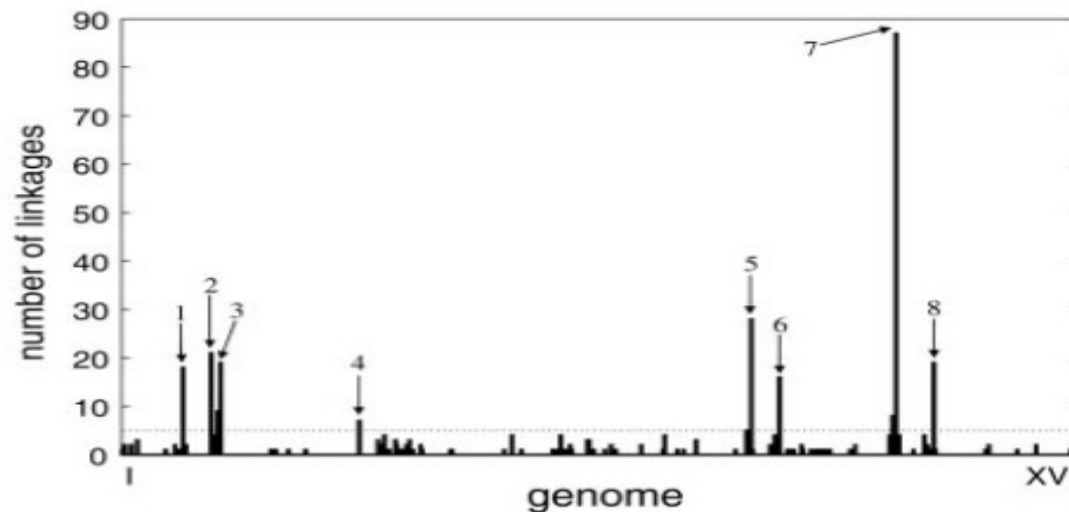
- Large heritability of transcripts abundance
- *Cis* eQTLs tend to have larger effects than *trans* eQTLs.



eQTL hot spots



Human dataset



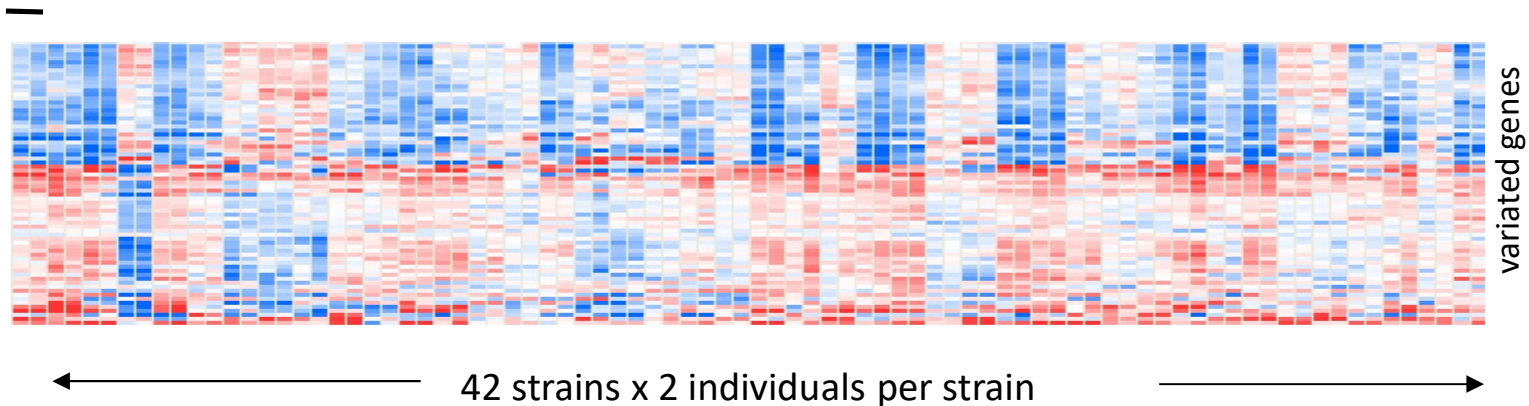
Yeast dataset

- Shared expression control regions suggest co-regulation

Morley et al. Nature 2004²⁸

The expression levels of many transcripts are highly correlated

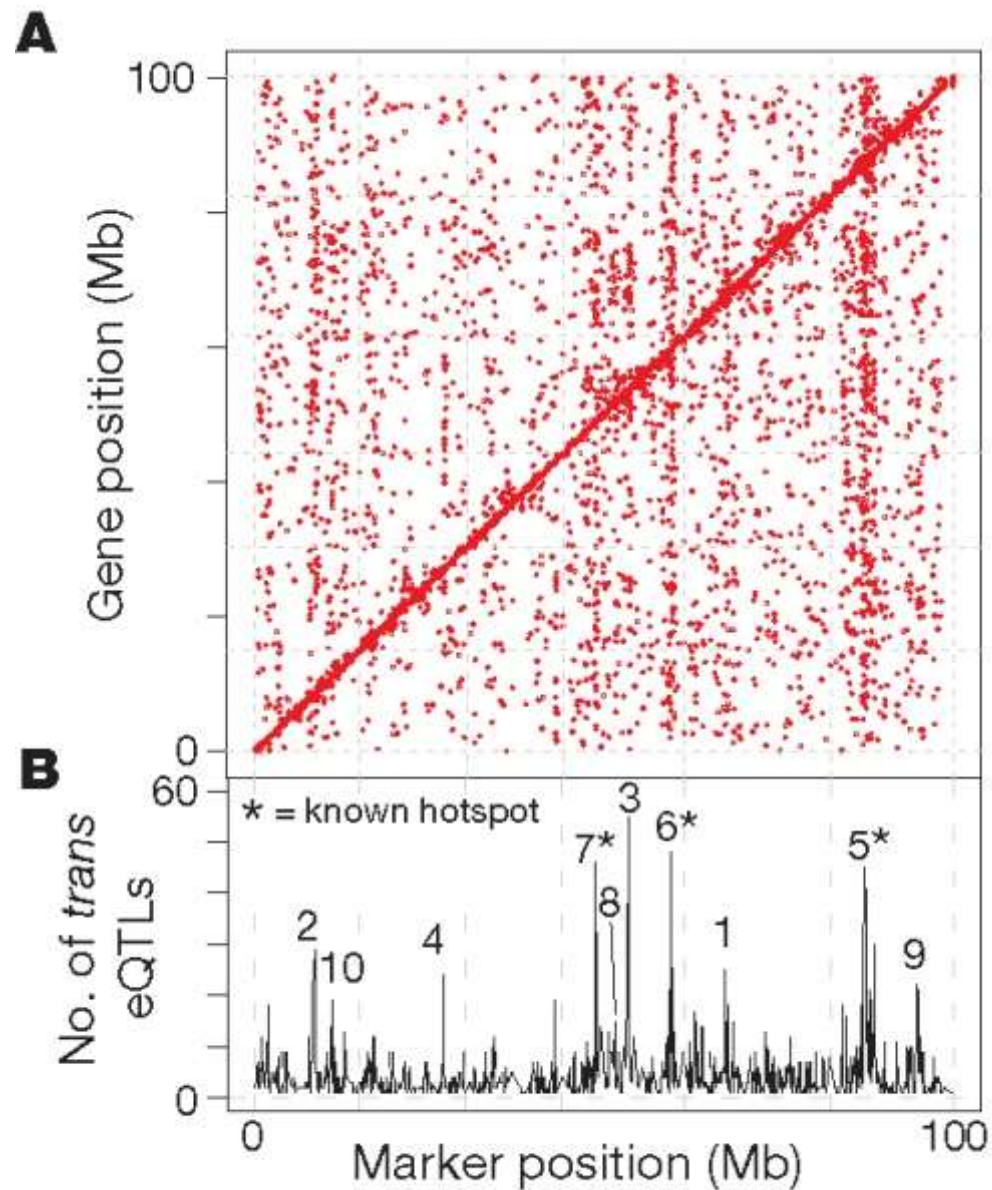
2 individual 1 strain



Common features of eQTL studies

- Large heritability of transcripts abundance
- *Cis* eQTLs tend to have larger effects than *trans* eQTLs.
- Some genomic regions are associated with variation in the expression levels of many transcripts (eQTL hot spots);
- The expression levels of many transcripts are highly correlated

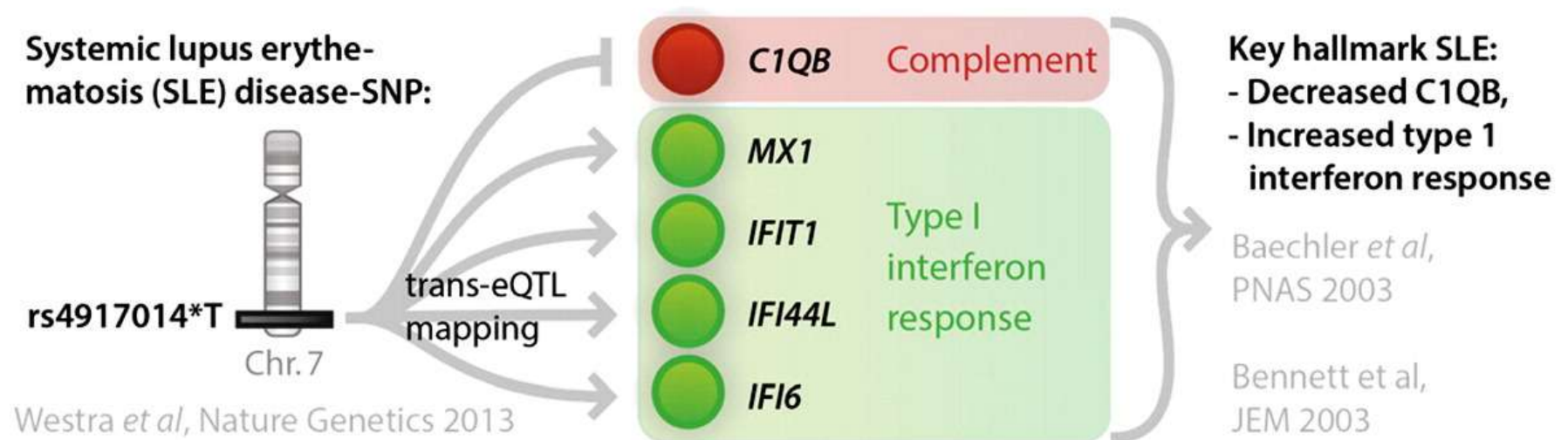
Summary visualization



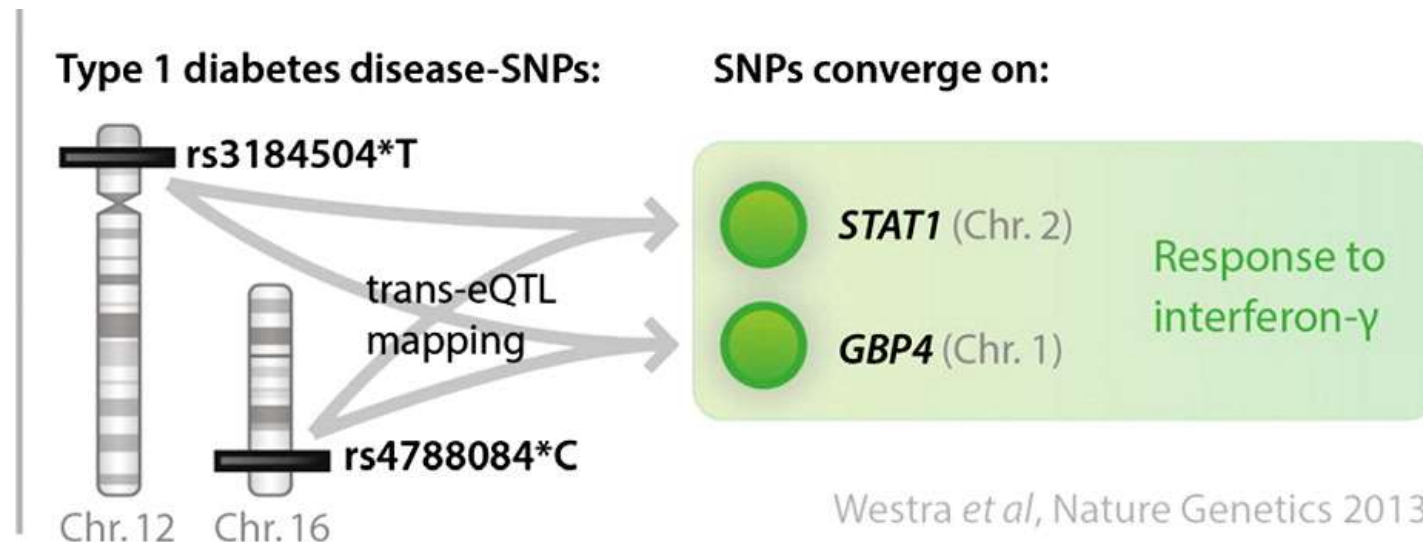
eQTL application: biological interpretation of GWAS data

Which genes are functionally implicated by a certain variant?

Integrating GWAS and functional genomics (trans-eQTLs) reveals hallmarks of disease



Integrating GWAS and functional genomics (trans-eQTLs) reveals key pathways in T1D



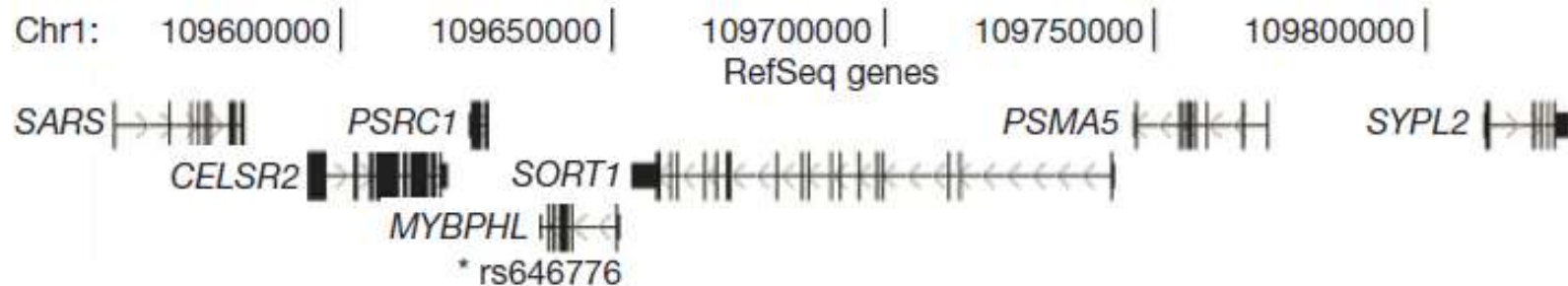
Risk factors for cardiovascular disease

Lead SNP	Lead trait	Other traits	Alleles/MAF	Effect size	P	eQTL
rs12027135	TC	LDL	T/A/0.45	-1.22	4×10^{-11}	Y
rs4660293	HDL		A/G/0.23	-0.48	4×10^{-10}	Y
rs2479409	LDL	TC	A/G/0.30	+2.01	2×10^{-28}	
rs2131925	TG	TC, LDL	T/G/0.32	-4.94	9×10^{-43}	Y
rs7515577	TC		A/C/0.21	-1.18	3×10^{-8}	
rs629301	LDL	TC	T/G/0.22	-5.65	1×10^{-170}	Y
rs1689800	HDL		A/G/0.35	-0.47	3×10^{-10}	
rs2642442	TC	LDL	T/C/0.32	-1.39	6×10^{-13}	
rs4846914	HDL	TG	A/G/0.40	-0.61	4×10^{-21}	
rs514230	TC	LDL	T/A/0.48	-1.36	5×10^{-14}	
rs1367117	LDL	TC	G/A/0.30	+4.05	4×10^{-114}	
rs1042034	TG	HDL	T/C/0.22	-5.99	1×10^{-45}	
rs1260326	TG	TC	C/T/0.41	+8.76	6×10^{-133}	Y
rs4299376	LDL	TC	T/G/0.30	+2.75	2×10^{-47}	
rs7570971	TC		C/A/0.34	+1.25	2×10^{-8}	
rs10195252	TG		T/C/0.40	-2.01	2×10^{-10}	Y
rs12328675	HDL		T/C/0.13	+0.68	3×10^{-10}	
rs2972146	HDL	TG	T/G/0.37	+0.46	3×10^{-9}	Y
rs2290159	TC		G/C/0.22	-1.42	4×10^{-9}	
rs645040	TG		T/G/0.22	-2.22	3×10^{-8}	
rs442177	TG		T/G/0.41	-2.25	9×10^{-12}	
rs13107325	HDL		C/T/0.07	-0.84	7×10^{-11}	Y
rs6450176	HDL		G/A/0.26	-0.49	5×10^{-8}	
rs9686661	TG		C/T/0.20	+2.57	1×10^{-10}	
rs12916	TC	LDL	T/C/0.39	+2.84	9×10^{-47}	
rs6882076	TC	LDL, TG	C/T/0.35	-1.98	7×10^{-28}	
rs3757354	LDL	TC	C/T/0.22	-1.43	1×10^{-11}	
rs1800562	LDL	TC	G/A/0.06	-2.22	6×10^{-10}	
rs3177928	TC	LDL	G/A/0.16	+2.31	4×10^{-19}	Y
rs2247056	TC		C/T/0.35	-2.00	2×10^{-15}	

Plasma concentrations of total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C) and triglycerides (TG).
Musunuru et al. Nature 2010
Teslovich et al. Nature 2010

The associated locus in 1p13

- GWASs have identified a locus on chromosome 1p13 strongly associated with both plasma low-density lipoprotein cholesterol (LDL-C) and myocardial infarction (MI – אוטם שריר הלב) in humans.
- This locus has the strongest association with LDL-C of any locus in the genome ($P < 10^{-170}$).

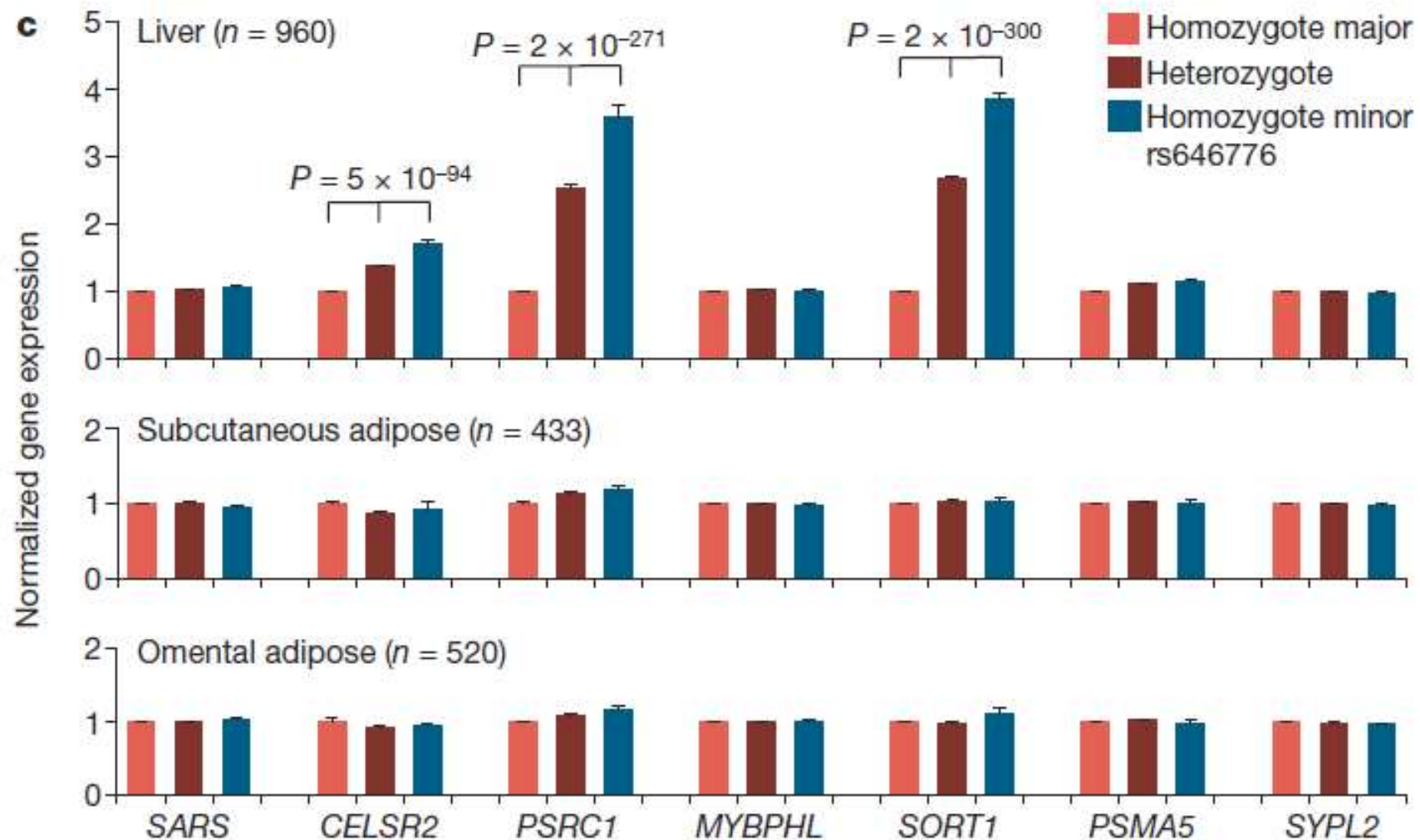


Goals:

To identify

- (1) the causal DNA variant in the 1p13 locus,
- (2) the gene regulated by the locus,
- (3) the mechanism by which the DNA variant affects the gene,
- (4) the mechanism by which the gene influences lipoprotein metabolism.

eQTL analysis



Hypothesis: variation in C/EBP activity affects SORT1 expression

Advantages:

- Relatively easy experimental phenotype (SORT1 expression in liver)
- Known tissue (liver)
- Cis hypothesis:

rs12740374

C/EBP consensus site

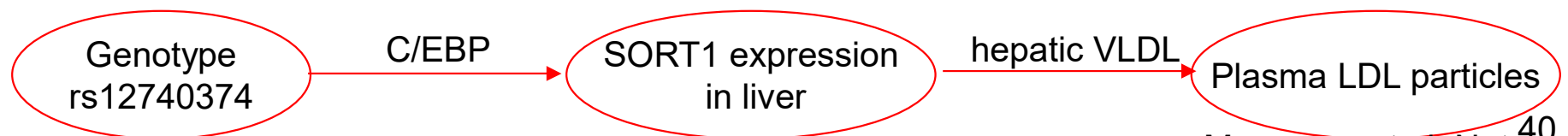
Minor TGGCTCGGCTGCCCTGAGGTTGCTCAATCAAGCACAGGT
1 2 3 4 5 6 7 8 9 10

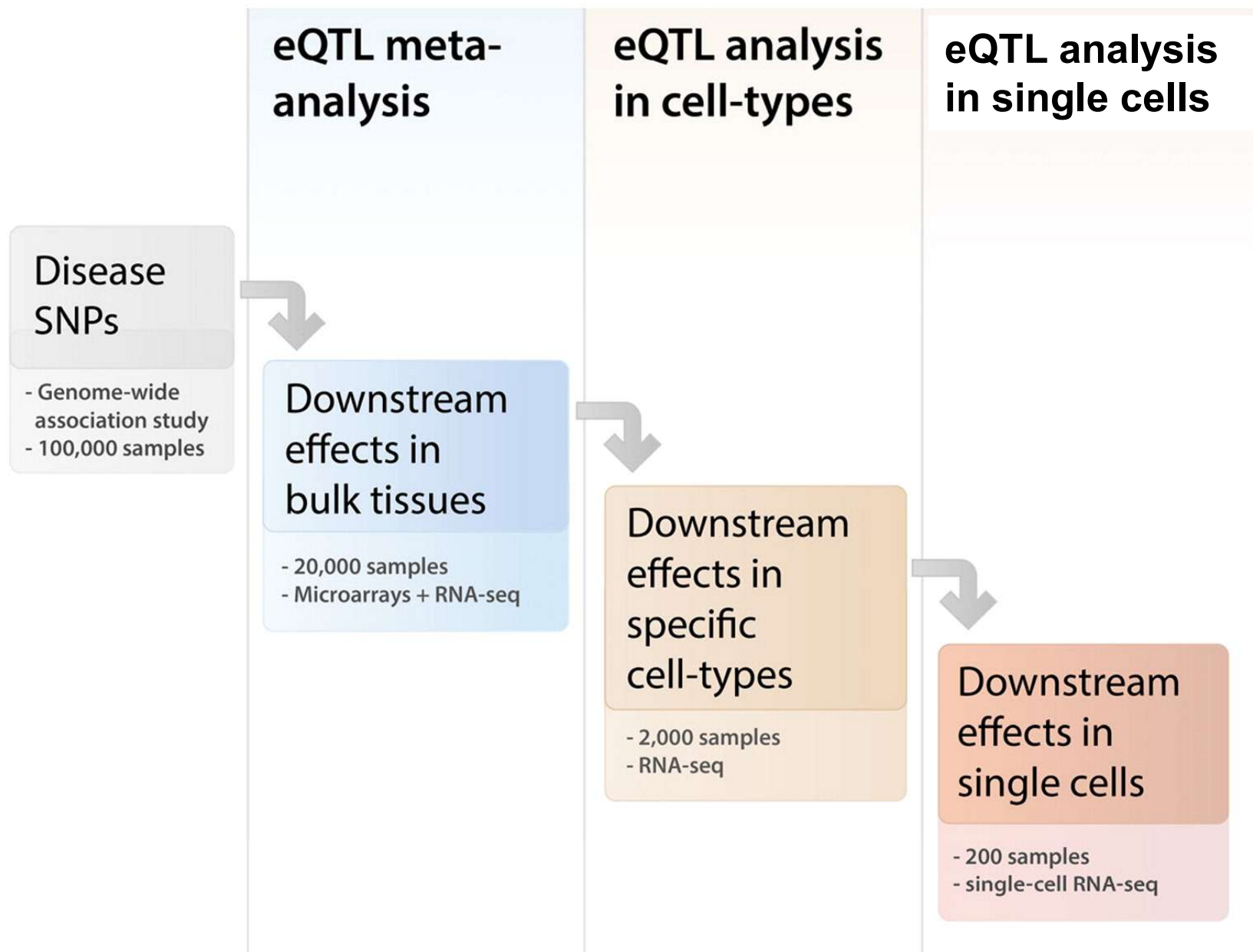
Major TGGCTCGGCTGCCCTGAGGCTGCTCAATCAAGCACAGGT

Mouse TGGCATGGTGGCCCTGAGGGGGC—CCCAGCACAGGT

Results in tissue culture: rs12740374, creates a C/EBP transcription factor binding site and alters the hepatic expression of the SORT1 gene.

Results using whole organism: Sort1 alters plasma LDL-C particle levels by modulating hepatic VLDL secretion.



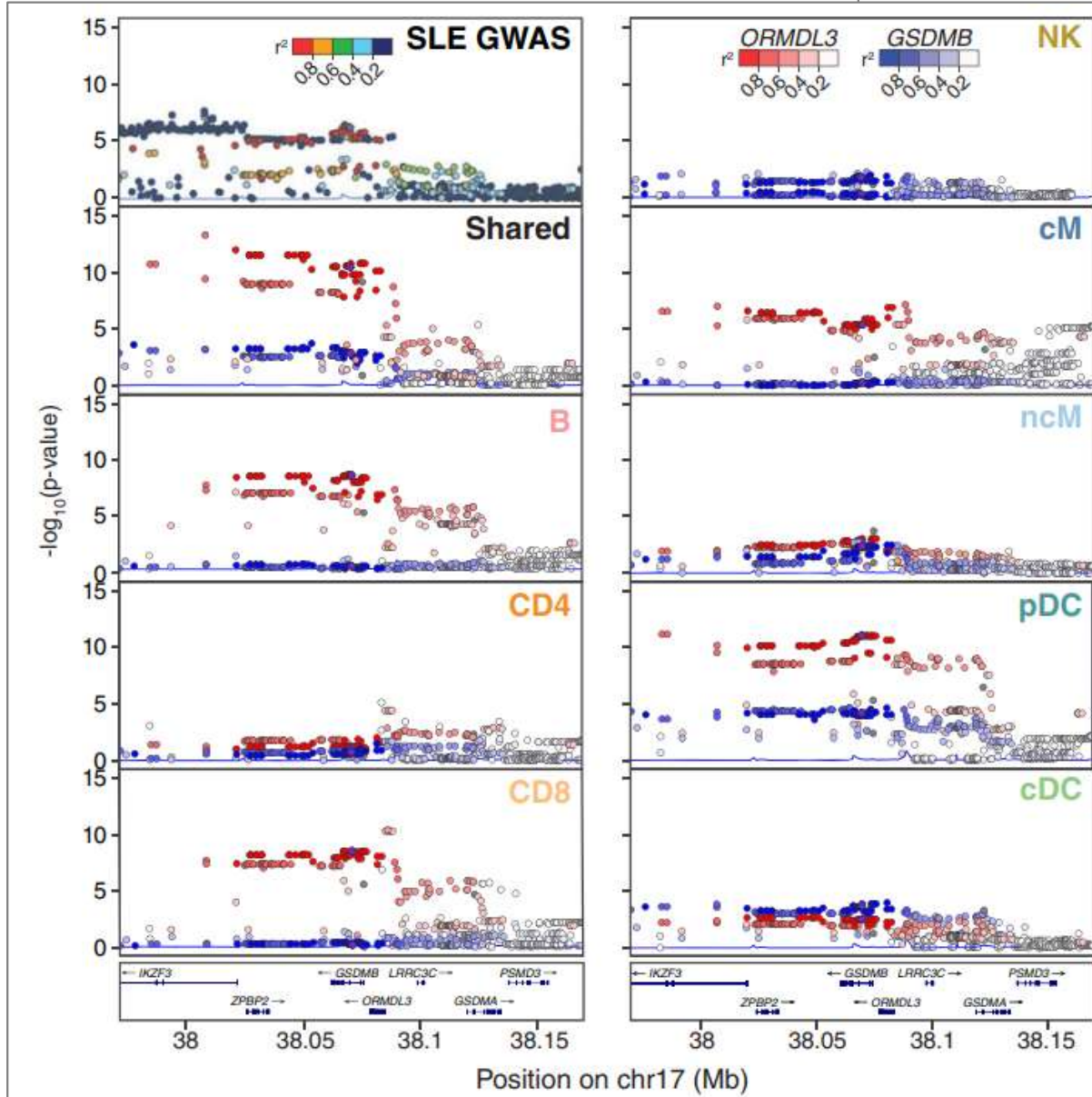


Example: lupus

Single-cell RNA-seq reveals cell type-specific molecular and genetic associations to lupus

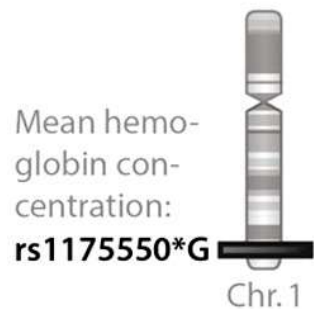
RICHARD K. PEREZ¹, M. GRACE GORDON¹, MEENA SUBRAMANIAM¹, MIN CHEOL KIM¹, GEORGE C. HARTOULAROS¹, SASHA TARG, YANG SUN¹, ANTON OGRODNIKOV, RAYMUND BUENO¹, [...], AND CHUN JIMMIE YE¹ +20 authors [Authors Info & Affiliations](#)

DOI: 10.1126/science.abf1970



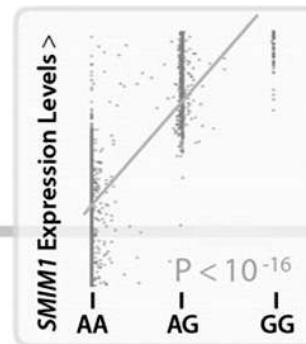
Integrating GWAS and functional genomics to reveal new hemoglobin-metabolism gene

GWAS on red blood cell traits:



Van der Harst *et al*, Nature 2012

Blood eQTL mapping:



Gene function prediction:

(GeneNetwork.nl, based on 80,000 RNA microarrays)

● Genes known to be involved in hemoglobin metabolism



Knock-down in zebrafish:

Reduced number of red blood cells

Cvejic *et al*, Nature Genetics 2013

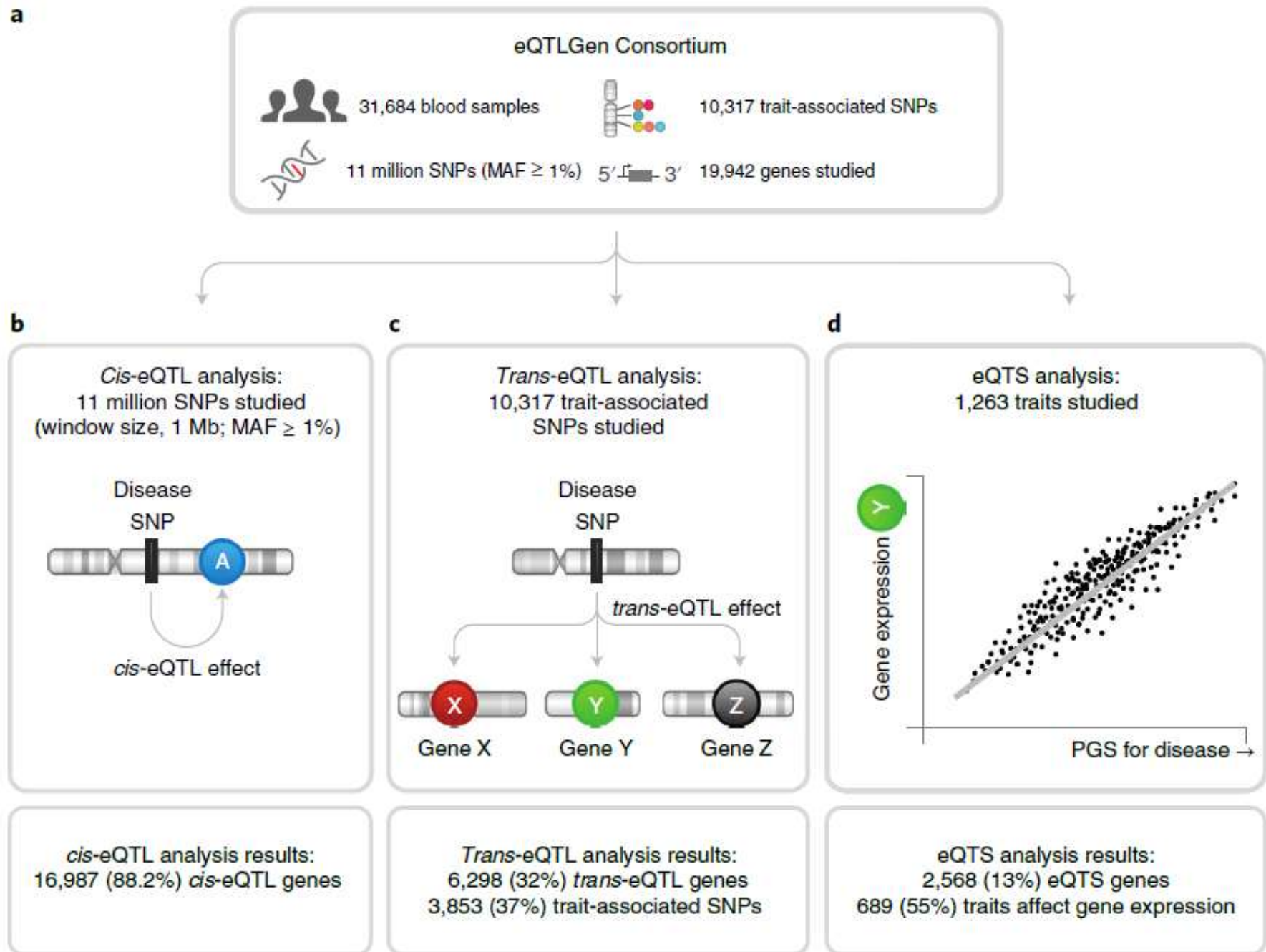


Large-scale *cis*- and *trans*-eQTL analyses identify thousands of genetic loci and polygenic scores that regulate blood gene expression

Urmo Vösa et al.

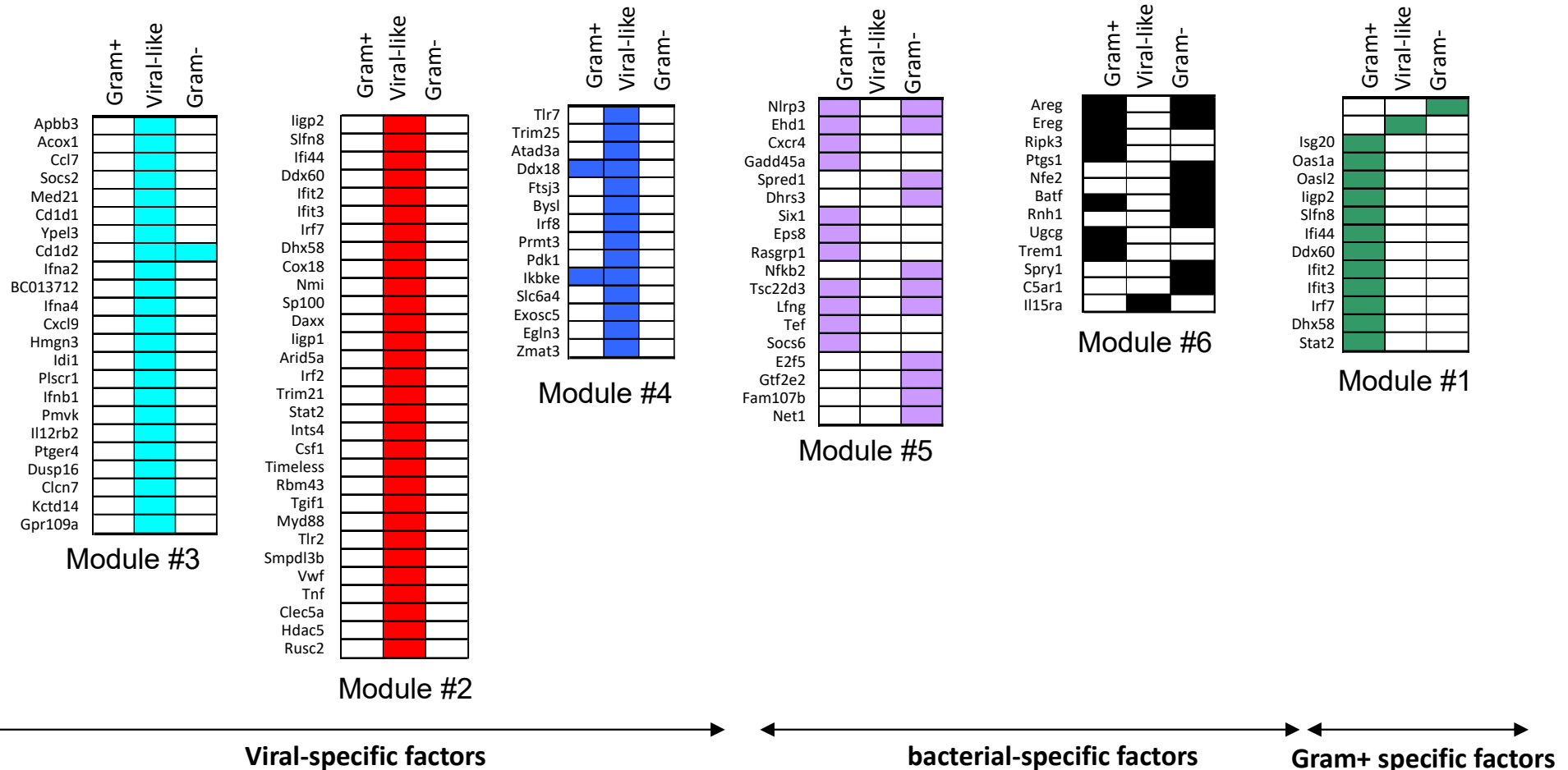
Nature Genetics | VOL 53 | September 2021 | 1300–1310 |

Overview of the study



eQTL modules in DCs response to pathogen

6 *trans*-acting (pathogen-specific) genetic factors



Pathogen-specific genetic factors may underlie the distinct variation in susceptibility to different pathogens

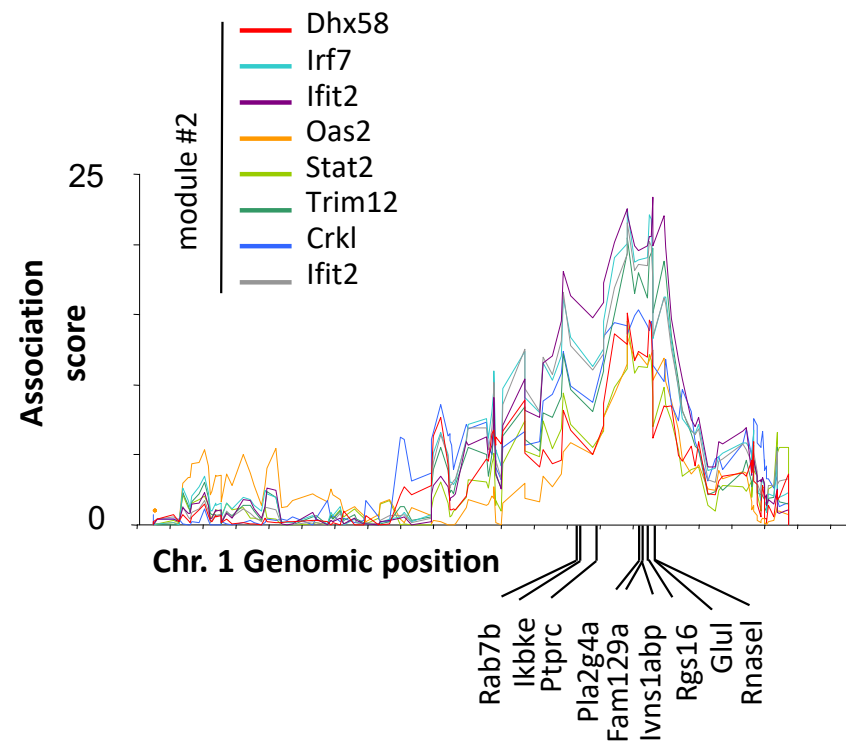
Module #2 is particularly intriguing

- It has the largest pleiotropic effect on 35 genes
- Its target genes are immune-related (e.g., TLR7, Dhx58, Stat2, Ifit2/3) and STAT and IRF targets ($P < 10^{-10}$)
- Affects its targets mainly in the context of viral stimulation

	Gram+	Viral-like	Gram-
Iigp2			
Sifn8			
Ifi44			
Ddx60			
Ifit2			
Ifit3			
Irf7			
Dhx58			
Cox18			
Nmi			
Sp100			
Daxx			
Iigp1			
Arid5a			
Irf2			
Trim21			
Stat2			
Ints4			
Csf1			
Timeless			
Rbm43			
Tgif1			
Myd88			
Tlr2			
Smpd13b			
Vwf			
Tnf			
Clec5a			
Hdac5			
Rusc2			

Module #2

The associated genomic interval at Chr1:155-165 Mb

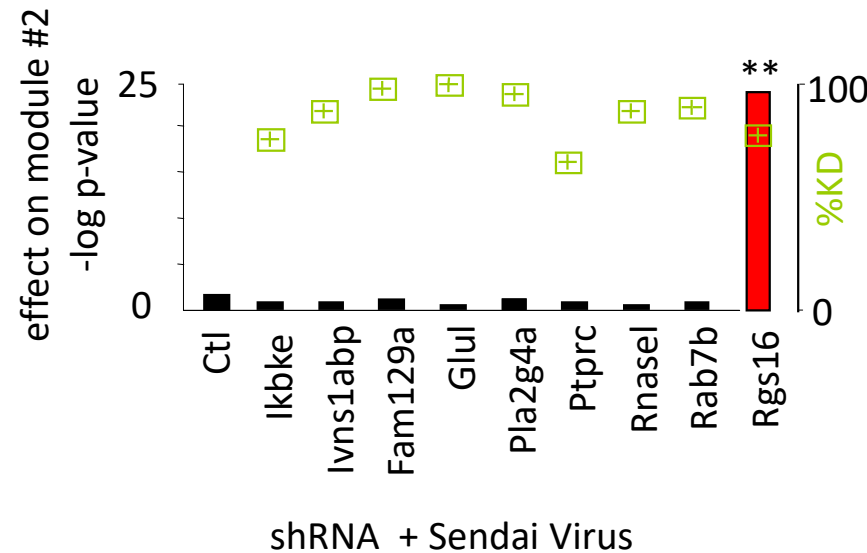


	Gram+	Viral-like	Gram-
ligp2			
Slfn8			
Ifi44			
Ddx60			
Ifit2			
Ifit3			
Irf7			
Dhx58			
Cox18			
Nmi			
Sp100			
Daxx			
Igip1			
Arid5a			
Irf2			
Trim21			
Stat2			
Ints4			
Csf1			
Timeless			
Rbm43			
Tgif1			
Myd88			
Tlr2			
Smpd13b			
Vwfr			
Tnf			
Clec5a			
Hdac5			
Rusc2			

Module #2

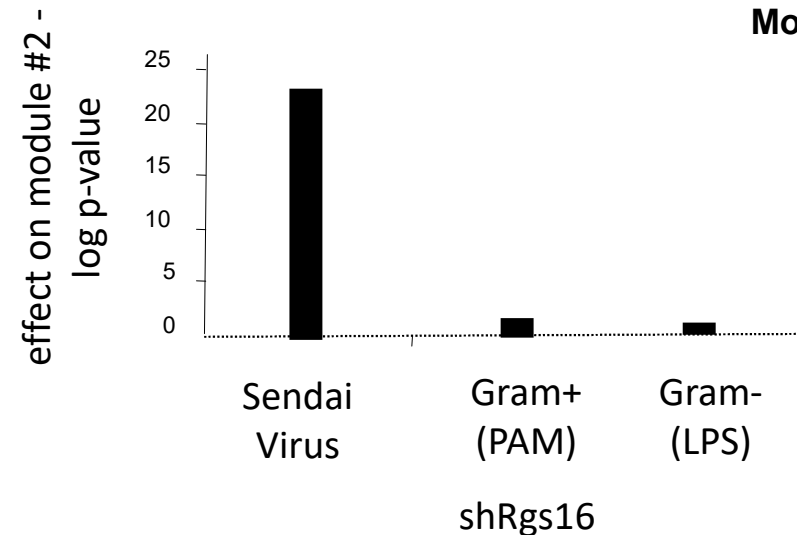
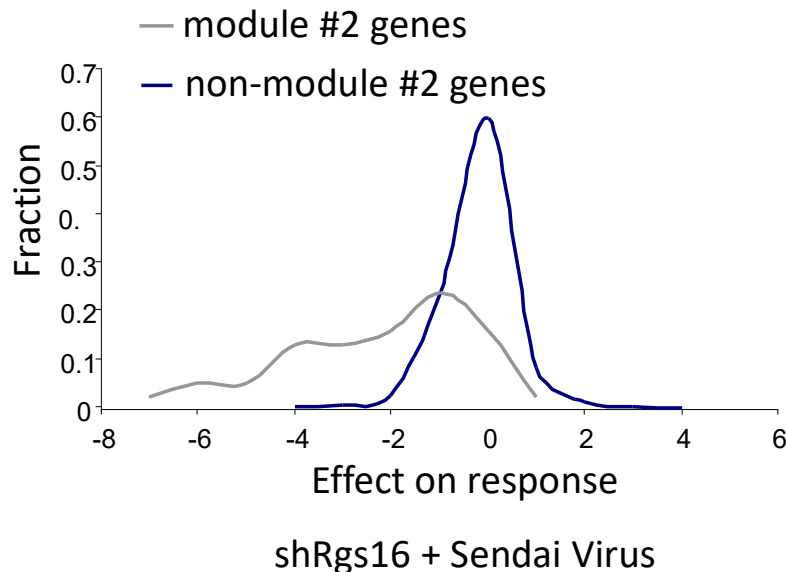
Regulator of G-protein signaling Rgs16: the likely underlying gene

Gene silencing through RNA interference

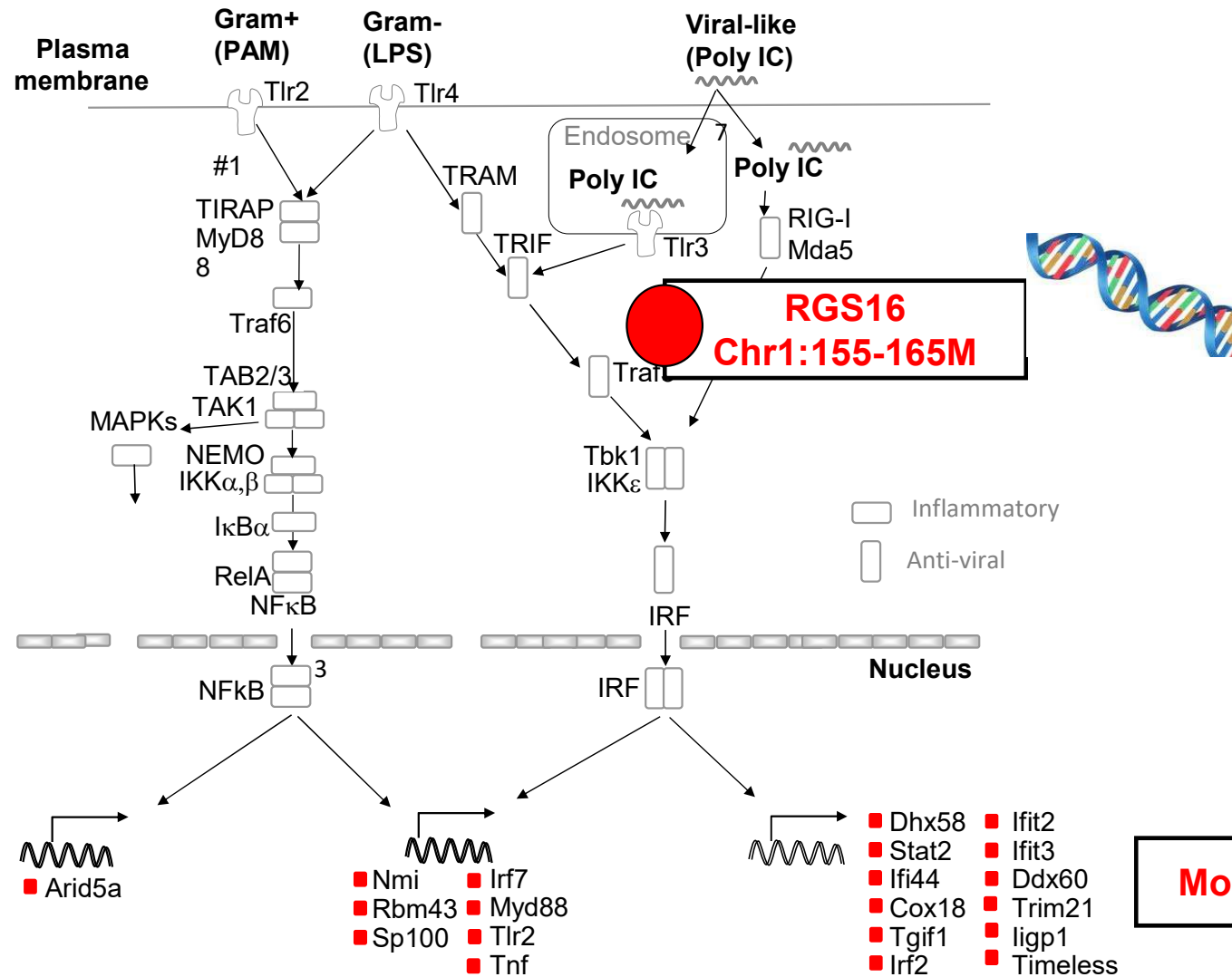


	Gram+	Viral-like	Gram-
ligp2			
Slfn8			
Ifi44			
Ddx60			
Ifit2			
Ifit3			
Irf7			
Dhx58			
Cox18			
Nmi			
Sp100			
Daxx			
Ilgp1			
Arid5a			
Irf2			
Trim21			
Stat2			
Ints4			
Csf1			
Timeless			
Rbm43			
Tgif1			
Myd88			
Tlr2			
Smpd13b			
Vwf			
Tnf			
Clec5a			
Hdac5			
Rusc2			

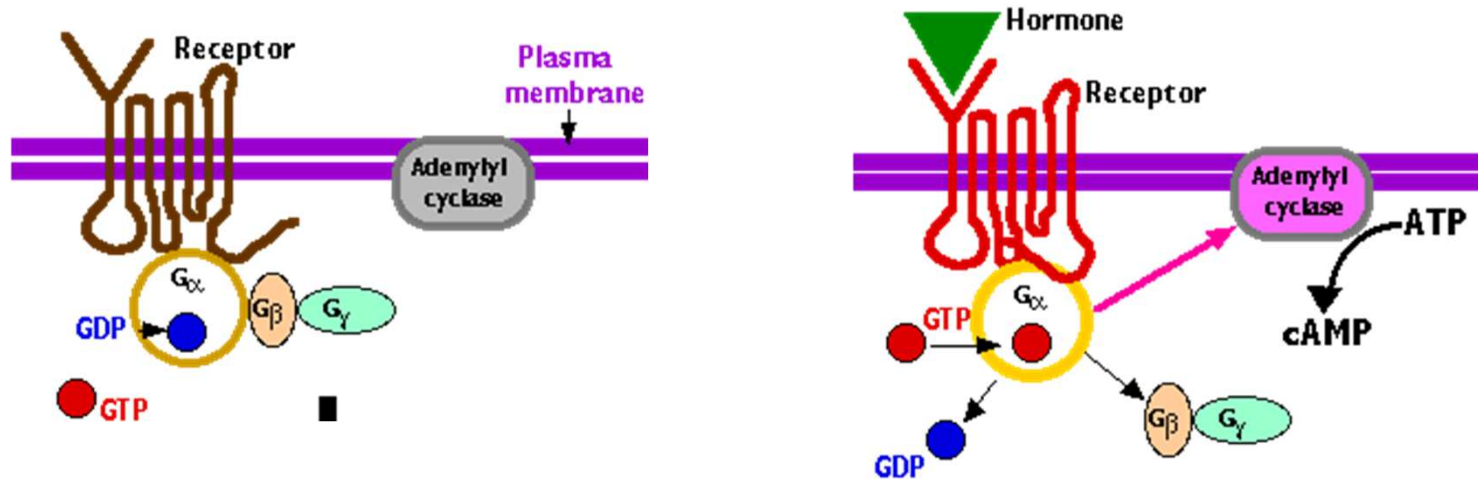
Module #2



A molecular model of gene-circuit-virus interaction



G proteins

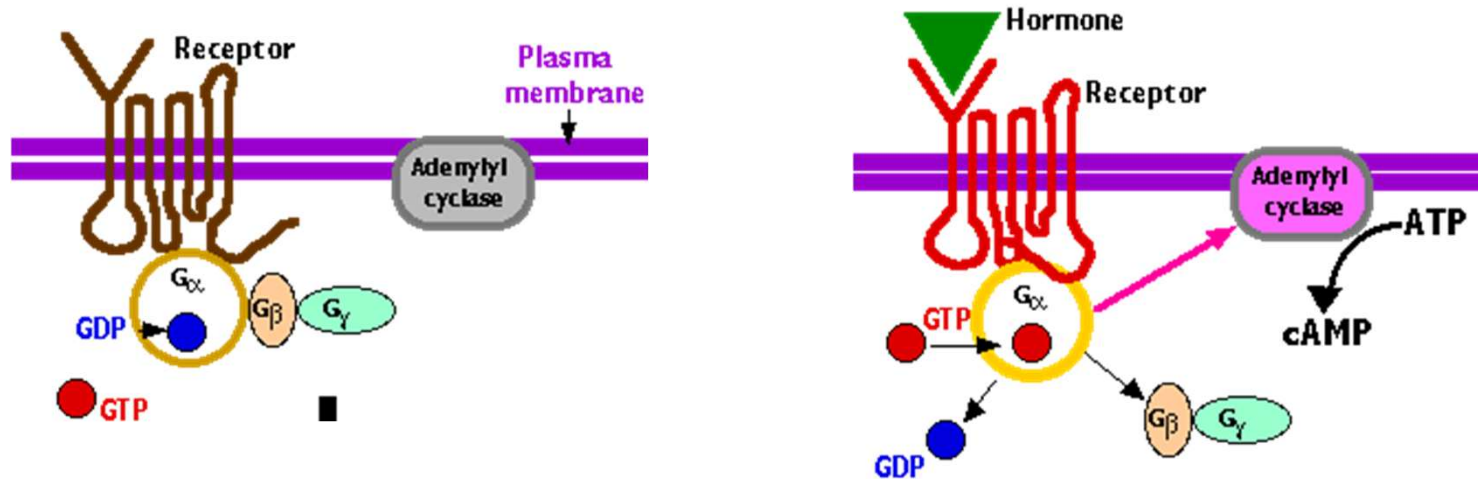


G proteins bind the guanine nucleotides GDP and GTP. They are associated with the inner surface of the plasma membrane and transmembrane 'G protein-coupled receptors' (GPCRs).

How They Work

- In the inactive state, G_{α} has GDP in its binding site.
- Ligand binding to the GPCR triggers GDP to leave and be replaced by GTP.
- GTP activates G_{α} , which in turn activates an effector molecule (such as adenyl cyclase).

G proteins



Activated G α is a GTPase so it quickly converts its GTP to GDP. This conversion restores the G protein to its inactive state.

- GTPase-activating proteins (GAPs) are a family of regulatory proteins whose members can bind to activated G proteins and stimulate their GTPase activity, with the result of terminating the signaling event. GAP are crucial in controlling the activity of G proteins.
- **GAP's role in this function is to turn the G protein's activity off.** In this sense, GAPs function is opposite to that of guanine nucleotide exchange factors (GEFs), which serve to enhance G protein signaling.
- **GAPs are also known as RGS protein** (Regulation of G proteins)

SNP rs32553675: a candidate causal mutation in Rgs16

